FluoroPolymers (FPs)

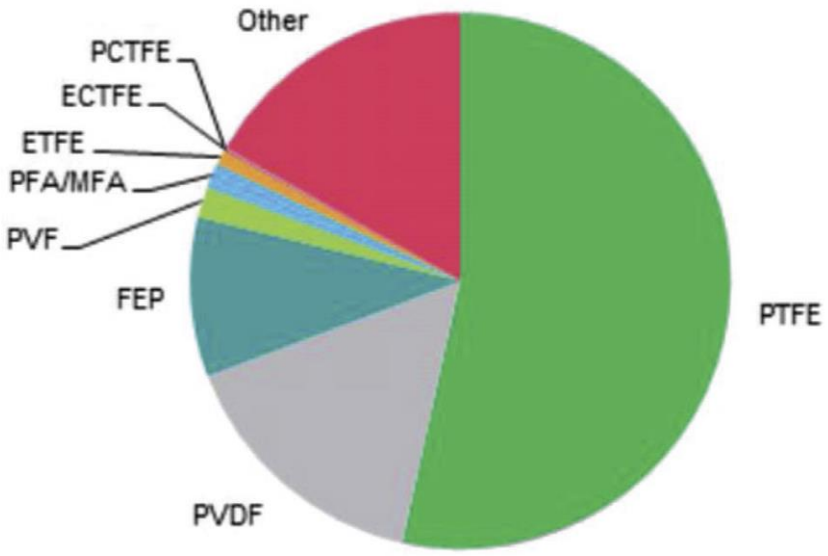

Figure 1. Global situation of the production of various FPs (2019).[16]

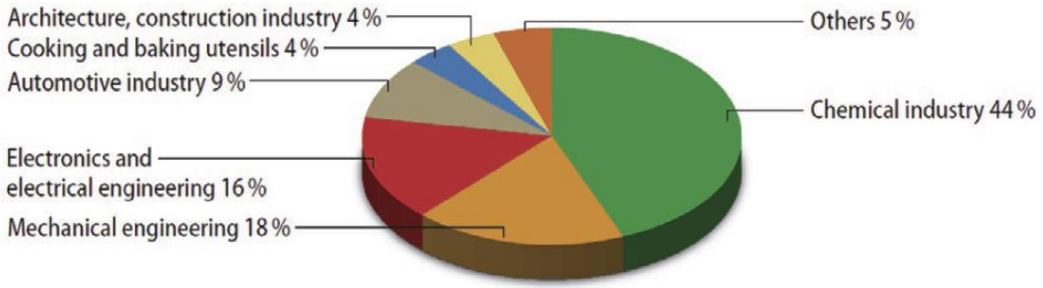

Figure 2. Fluoropolymers consumption per industry in Europe (2015). Reproduced with permission.[17] Copyright 2016, Carl Hanser Verlag GmbH & Co.

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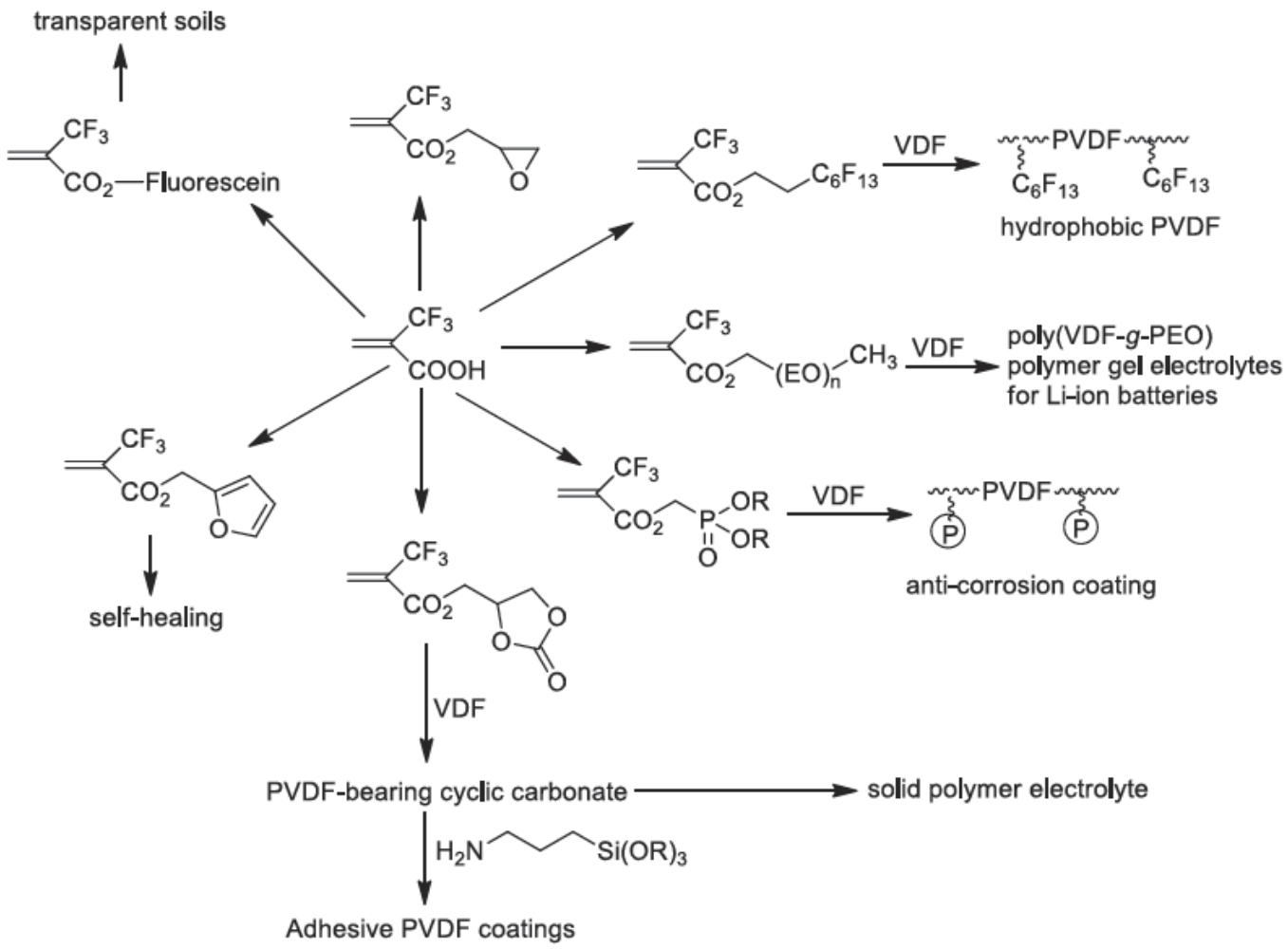
FluoroPolymers (FPs)



Scheme 2. Synthesis of various functional 2-trifluoromethacrylates (MAF-esters) from 2-trifluoromethacrylic acid and their radical copolymerizations with VDF for specific materials.[25]

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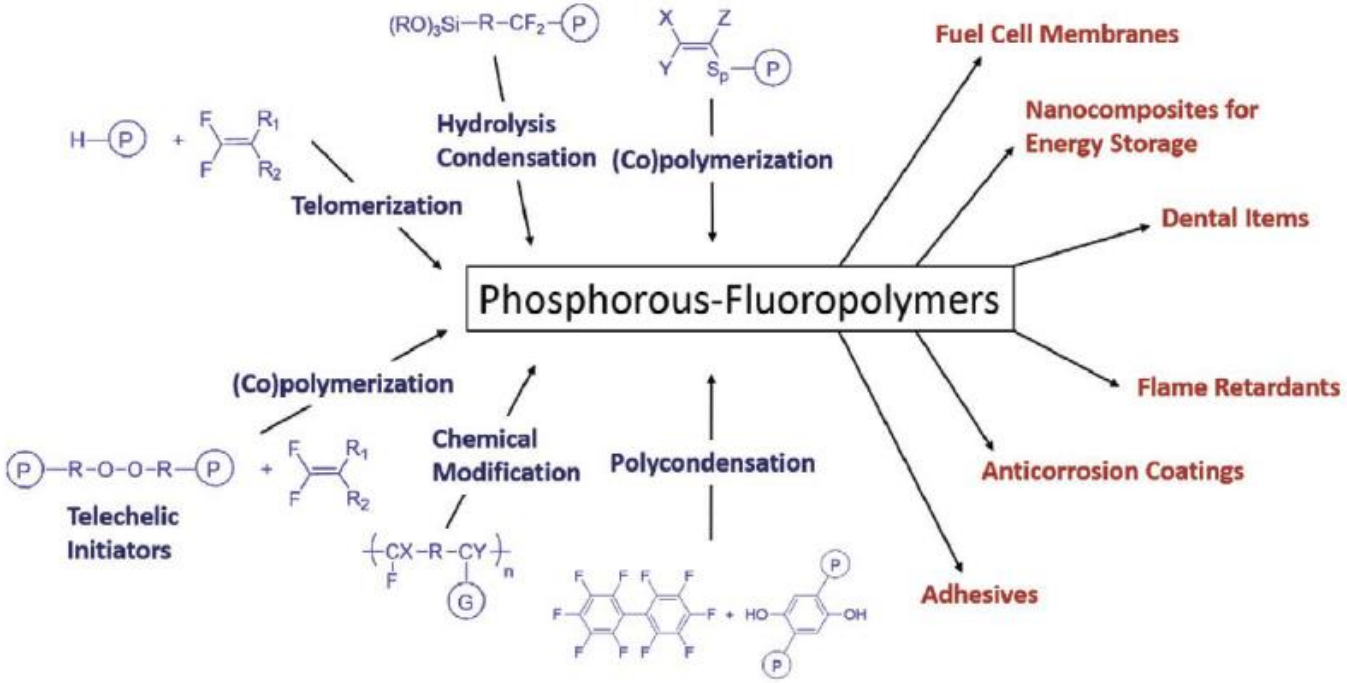
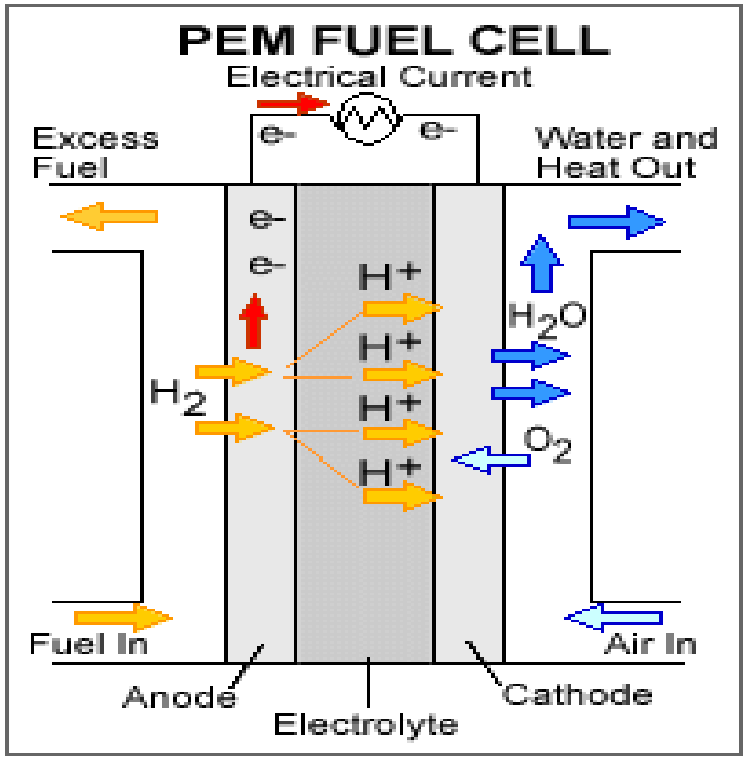


Scheme 3. Strategies of syntheses and applications of phosphorous-containing fluoropolymers.[30]

proton exchange membrane fuel cells
(PEM fuel cells,or PEMFC)



a proton-conducting polymer membrane (Eg: nafion) contains the electrolyte solution that separates the anode and cathode sides.
This was called a "solid polymer electrolyte fuel cell" (SPEFC)

Proton exchange membrane fuel cell

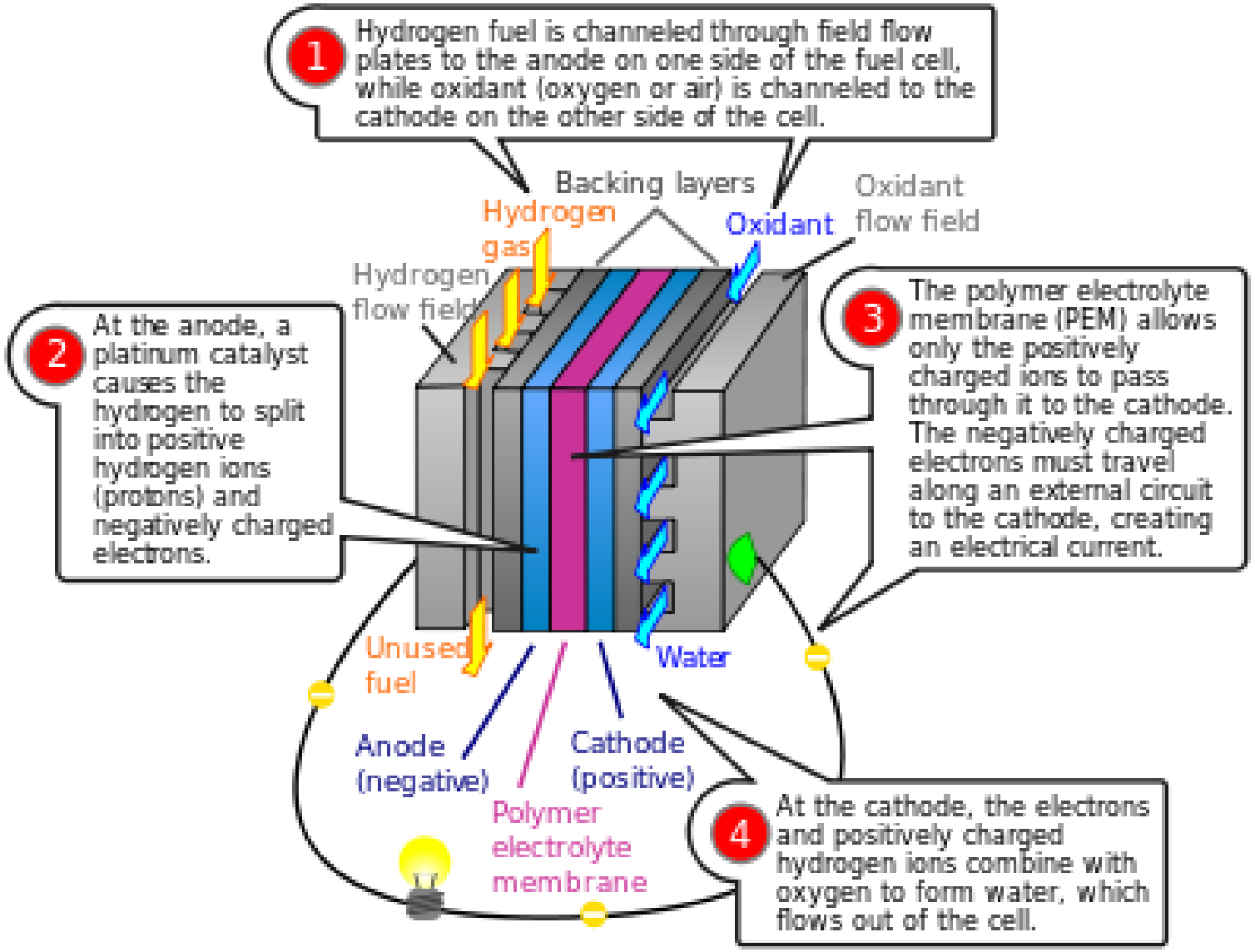


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Table 2. Major domains, properties, and industrial applications from fluoropolymers.[18]

Industry/application domains	Searched properties	Specific uses	Used fluoropolymers[a)]
Chemical/etrochemical industry	Chemical resistance Good mechanical properties Thermal stability Cryogenic properties	Gaskets, vessel liners, pumps, valve and pipe liners, tubings, coatings, expansion joints/bellows, heat exchangers	PTFE, PFA/MFA ETFE, ECTFE FEP FKM, FFKM TFE-P
Electrical/electronic industry	Low dielectric constant High volume/surface resistivity High dielectric breakdown voltage Flame resistance, thermal stability Low refractive indices	Wire and cable Insulation, connectors, optical fibers, printed circuit boards	FEP, PTFE, PFA, MFA ETFE, ECTFE PCTFE amorphous FP
Automotive/aircraft industry	Low coefficient of friction Good mechanical properties Cryogenic properties, Chemical resistance Low permeation properties	Seals, O-Rings, hoses in automotive power steering, transmissions, and airconditioning, bearings, sensors fuel management systems.	FKM, PTFE FFKM THV
Coatings	Thermal/weather stability Low surface energy chemical resistance	Cookware coatings, coatings of metal surfaces,***powder coatings	PTFE PVDF, ETFE FEVE, PFA
Medical	Low surface energy, Stability, purity Excellent mechanical properties Chemical resistance	Cardiovascular grafts, heart patches, ligament replacement packaging films for medical products	PTFE, PCTFE
General Architectural/fabric/film applications	Excellent weatherability Flame resistance Transparency Low surface energy Barrier properties	Coated fabrics and films for buildings/ roofs, front/backside films for solar applications	ETFE, PTFE, PVDF PCTFE, PVF, THV

FluoroPolymers (FPs)

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Table 2. Major domains, properties, and industrial applications from fluoropolymers.[18]

Industry/application domains	Searched properties	Specific uses	Used fluoropolymers[a)]
	Barrier properties		
Polymer additives	Low coefficient of friction Flame resistance Abrasion resistance Antistick properties	Polyolefin processing to avoid surface defects and for faster processing. Additives for inks, coatings, lubricants, antidripping agents	THV, FKM PVDF, PTFE
Semiconductor industry	Chemical resistance High purity Antiadhesion, insulation, barrier properties Thermal stability	Process surfaces Wafer carriers Tubing, valves, pumps and fittings, Storage tanks	PFA, ECTFE PCTFE, PTFE Amorphous FP
Energy conversion/storage Renewable energies	Chemical/thermal resistance Ion-transportation High weatherability High transparency Corrosion resistance	Binder for electrodes, separators, Ion-selective membranes, gaskets, membrane-reinforcements, Films for photovoltaics coatings for wind mill blades	PVDF, Fluoroionomers (PFSA),THV, ETFE ECTFE, PTFE, FEP PVF

[a)]Tetrafluoroethylene (TFE), vinylidene fluoride (VDF), chlorotrifluoroethylene (CTFE), 3,3,3-trifluoropropene (TFP), hexafluoropropylene (HFP), trifluoroethylene (TrFE), perfluoromethyl vinyl ether (PMVE), ethylene (E), perfluorosulfonic acid (PFSA), poly(VDF-*ter*-HFP-*ter*-TFE) terpolymers (THV).

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